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Compilation Mode

Action	Key binding	Command	Note
Compilation-Mode ○ Help • Compile • Navigate in compilation log buffer • next-error-follow-minor-mode • Navigate to next/previous error • kill compilation, quit buffer ○ PEL controlled save buffer mgt Last updated on:	Emacs provides a special mode to display the result of compilation and analyzer commands that list errors and warnings detected on specific lines and columns: the **compilation-mode**. • Some major modes provide another mode derived from the compilation-mode that is specialized for the programming language supported by the mode. • For example, emacs-lisp-mode uses the emacs-lisp-compilation-mode. • These derived modes normally provide the same commands and key bindings. Any difference is described in the page for the mode. In all cases you can always request information for the mode by typing **C-h m** or **<f1> m** in the window using the compilation-mode or its derived counterpart. **★★PEL controlled buffer save management** In several major modes, the compile command prompts to save each buffer currently modified, it prompts for the buffer visiting the file being compiled but all other buffers as well. This can be quite annoying. PEL provides a controlled buffer save management workspace that reduces the prompting and provides specialized buffer to quickly identify which buffers need to be saved before proceeding with the compilation or the file analysis. 2026-10-27	See also: **Hide/Show**	
Open this PDF file See also: **Help/Info**	**<f12> <f1>**	(**pel-help-pdf** &optional OPEN-WEB-PAGE)	From a compilation-mode or coming-mode buffer: open the **Compilation Mode** local PDF. With **C-u** , **M--**, **M- -1** or **C- -1** prefix: open the remote GitHub hosted PDF instead. If the **pel-flip-help-pdf-arg** user-option is set it's the other way around.
Show info about the PEL controlled buffer save management for compile command	**<f12> <f2>**	(**pel-compile-info**)	From a compilation-mode or coming-mode buffer open the special Help buffer that describes the specialized ibuffer prompt used by PEL to let you quickly identify buffers that must be saved before performing the compilation or the file analysis operation. Shows current values and provides buttons to quickly access the customization buffers of the relevant user-options.
Customize compilation	**<f12> <f3>**	(**pel-compile-customize**)	From a compilation-mode or coming-mode buffer, open a buffer to browse the compilation customization group hierarchy with Emacs customization tree browser. Use this to access every user-option that controls the compilation mechanisms.
To Compile	Several major modes have their own command to launch the compilation. • The compilation commands are described in the page describing the major mode. • PEL tries to bind the command for compilation to the **<f12> c** key sequence. In some cases that is a prefix to more commands related to compilation.		
Compile	**M-x compile**		Refer to the major mode for the compilation command specific to the mode.
compilation-mode buffer commands	The following commands are issued from the compilation-mode log buffer that opens when the compilation command detected errors and or warnings.		
Move to beginning of buffer See also: **Navigation**	• **<** • **M-<** • **C-<home>**	(**beginning-of-buffer** &optional ARG)	Move point to the beginning of the buffer. • With numeric arg N, put point N/10 of the way from the beginning. • If the buffer is narrowed, this command uses the beginning of the accessible part of the buffer. • Push mark at previous position, unless either a **C-u** prefix is supplied, or Transient Mark mode is enabled and the mark is active.
Move to end of buffer See also: **Navigation**	**>**	(**end-of-buffer** &optional ARG)	Move point to the end of the buffer. • With numeric arg N, put point N/10 of the way from the end. • If the buffer is narrowed, this command uses the end of the accessible part of the buffer. • Push mark at previous position, unless either a **C-u** prefix is supplied, or Transient Mark mode is enabled and the mark is active.
Toggle next-error-follow-minor-mode When this mode is active, navigating across errors in the compilation log buffer causes automatic display of the faulty line in the source code window.	**C-c C-f**	(**next-error-follow-minor-mode** &optional ARG)	Toggles next-error-follow-minor-mode for compilation, occur and diff modes. • If the prefix argument is positive, enable the mode, and if it is zero or negative, disable the mode. • When turned on, cursor motion in the compilation, grep, occur or diff buffer causes automatic display of the corresponding source code location. To check whether the minor mode is enabled in the current buffer, type the following key sequence: **M-: next-error-follow-minor-mode <RET>**
Display the source of current error inside another window	**C-o**	(**compilation-display-error**)	Display the source for current error in another window. • Useful if the compilation-mode log buffer uses the complete Emacs frame.
Visit source file for current error message	**C-c C-c**	(**compile-goto-error** &optional EVENT)	Visit the source code for the current error message: move point inside the source code file at the line and column identified by the message.
Next error no select (Emacs >= 28.1)	**n**	(**next-error-no-select** &optional N)	Move point to the next error in the 'next-error' buffer and highlight match. • Prefix arg N says how many error messages to move forwards (or backwards, if negative). Finds and highlights the source line like **C-x `**, but does not select the source buffer.
Compilation next error	**M-n**	(**compilation-next-error** N &optional DIFFERENT-FILE PT)	Move point to the next error in the compilation buffer. • Prefix arg N says how many error messages to move forwards (or backwards, if negative). This does NOT find the source line like **C-x `**.
Next Error and visit file ☝In most major modes this can be issued from the source file buffer.	• **C-x `** • **M-g n** • **M-g M-n**	(**next-error** &optional ARG RESET)	Move to the next error message and visit the corresponding source code. • If all the error messages parsed so far have been processed already, the message buffer is checked for new ones. • A prefix ARG specifies how many error messages to move; negative means move back to previous error messages. • Just **C-u** as a prefix means reparse the error message buffer and start at the first error.
Previous error no select (Emacs >= 28.1)	**p**	(**previous-error-no-select** &optional N)	Move point to the previous error in the 'next-error' buffer and highlight match. • Prefix arg N says how many error messages to move backwards (or forwards, if negative). Finds and highlights the source line like **M-g p**, but does not select the source buffer.
Compilation previous error	**M-p**	(**compilation-previous-error** N)	Move point to the previous error in the compilation buffer. • Prefix arg N says how many error messages to move backwards (or forwards, if negative). This does NOT find the source line like **M-g p**.
Previous Error and visit file ☝In most major modes this can be issued from the source file buffer.	• **M-g p** • **M-g M-p**	(**previous-error** &optional N)	Visit previous previous error message and corresponding source code. • Prefix arg N says how many error messages to move backwards (or forwards, if negative).
Compilation error on previous file	**M-{**	(**compilation-previous-file** N)	Move point to the previous error for a different file than the current one. Prefix arg N says how many files to move backwards (or forwards, if negative).
Compilation error on next file	**M-}**	(**compilation-next-file** N)	Move point to the next error for a different file than the current one. Prefix arg N says how many files to move forwards (or backwards, if negative).
Quit window	• **q** • **C-x w q**	(**quit-window** &optional KILL WINDOW)	Quit WINDOW and bury its buffer. • With prefix argument KILL non-nil, kill the buffer instead of burying it.
Kill compilation	**C-c C-k**	(**kill-compilation**)	Kill the compilation process.

Action	Key binding	Command	Note
PEL controlled buffer save management			When issuing a compile command for some major mode, PEL takes over controlling which buffers may need to be saved and how they are saved. ● **pel-compile-save-mode-scope** : identifies the modified buffer candidate selection policy. The candidates are selected by one of the following policies: • `Emacs default`: all modified buffers • `same-mode`: all modified buffers of the same mode as buffer rising compiled/analyzed file (in either classic or Tree-Sitter major mode), • `related`: all modified buffers of the same mode as buffer rising compiled/analyzed file or identified related modes (in either classic or Tree-Sitter major mode). The related modes are identified by the **pel-compile-related-modes-alist** user-option. • `none`: None of the other modified buffers is saved. The only candidate is the buffer visiting the file being compiled/analyzed. ● The prompt mechanism is selected when you issue the compilation command and several modified buffers are identified as save candidates by the currently used buffer save policy. It also depends on whether the number of modified buffer candidates is larger than the value identified by the **pel-compile-batch-save-threshold** user-option. • If # modified buffer candidates < **pel-compile-batch-save-threshold**: PEL prompts for each buffer sequentially, as Emacs would normally do. • If # modified buffer candidates ≥ **pel-compile-batch-save-threshold:** PEL first prompts for the selection method and offer the following methods: • **s**: Sequential prompting (like Emacs default) • **i**: Listing all modified buffers that may need saving in a specialized IBuffer where you can identify the buffers that actually need to be saved and they are all saved in batch, the default list all buffers and you can easily select all buffers to save. Once the selection is done proceed by typing **x** to execute the compilation command. • **k**: Skip saving; save no buffer, and compile, • **q**: quit/abort; no compilation is done. ● The specialized IBuffer with **pel--compile-save-workspace-mode** minor mode.

• Prompts for save selection method			When the number of candidate modified buffers equals or exceeds the value identified by **pel-compile-batch-save-threshold**, PEL prompts for the selection method:
			`PEL: 16 modified buffers found.`
			`[i] Show in Ibuffer, [s] Sequential prompt, [k] Skip saves, [q] Quit/Abort:`

• The specialized Ibuffer shows the names of all relevant modified buffers.
• At first they are all selected: they will all be saved if you type **x** to execute the compilation/analysis command.
• Unmark one buffer by moving to the line and type **u** key, or unmark all buffers by typing the **U** key.
• Type the **m** key to mark a buffer again if you decide it needs to be saved.
• Type **d** key to show the diff between the buffer and the file it visits, **D** to compare the file with its VCS version.
• Type **x** to save marked buffers and execute the compilation/ analysis command.
• Type **q** to abort.
• Type **?** to execute **pel-compile-info**: open a specialized help buffer to show more info on the mode and its customization.
• You can also use the other ibuffer commands to sort and filter the list.
			If you select **i** to show the list in Ibuffer, then PEL opens the specialized Ibuffer to list all modified buffers that may need to be saved:
			`Save before compile? | m: save | u/U: no save/all| d: Δbuf/file | D: Δfile/VC | x: compile | q: abort | ?: help | (FILTERS active)`
			`MRL Name Size Mode Filename/Process`
			`[ Compilation Mode Targets ]`
			`S* build-emacs-fro... 10490 ReST ~/my/dvp/elisp/pel/doc/build-emacs-from-source.rst.txt`
			`S* comparing-pel-t... 23095 ReST ~/my/dvp/elisp/pel/doc/comparing-pel-to-others.rst.txt`
			`S* creating-extens... 4552 ReST ~/my/dvp/elisp/pel/doc/creating-extension-less-scripts.rst.txt`
			`S* cross-reference... 6287 ReST ~/my/dvp/elisp/pel/doc/cross-reference-with-ctags.rst.txt`
			`S* emacs-daemon.rs... 37494 ReST ~/my/dvp/elisp/pel/doc/emacs-daemon.rst.txt`
			`S* using-tree-sitt... 14399 ReST ~/my/dvp/elisp/pel/doc/using-tree-sitter.rst.txt`
			`S* file-directory-... 3810 ReST ~/my/dvp/elisp/pel/doc/file-directory-operations.rst.txt`
			`S* install_e_ge_ec... 3292 ReST ~/my/dvp/elisp/pel/doc/install_e_ge_ec.rst.txt`
			`S* pel-ffind.rst.txt 1230 ReST ~/my/dvp/elisp/pel/doc/pel-ffind.rst.txt`
			`S* pel-indentation... 16467 ReST ~/my/dvp/elisp/pel/doc/pel-indentation-tab-control-setup.rst.txt`
			`S* pel-major-mode-... 5886 ReST ~/my/dvp/elisp/pel/doc/pel-major-mode-setup.rst.txt`
			`S* pel-startup-tim... 2915 ReST ~/my/dvp/elisp/pel/doc/pel-startup-time.rst.txt`
			`S* project-time-tr... 16647 ReST ~/my/dvp/elisp/pel/doc/project-time-tracking-with-org.rst.txt`
			`S* tree-sitter-gra... 3002 ReST ~/my/dvp/elisp/pel/doc/tree-sitter-grammars.rst.txt`
			`S* upgrading-emacs... 10045 ReST ~/my/dvp/elisp/pel/doc/upgrading-emacs.rst.txt`
			`S* using-emacs-sou... 3087 ReST ~/my/dvp/elisp/pel/doc/using-emacs-source.rst.txt`
			`16 buffers 162698 16 files, no processes`
			`-UUU:%*- F1 *PEL Compile Save Workspace* All (3,0) (IBuffer by recency (Auto) WK PEL-Save) 08:57 3.34 ----------`
			`PEL: [m] Mark Save | [u/U] Unmark | [d] Diff | [D] VCS Diff | [x] Save & Compile | [?] help`
			
			See ∑ ibuffer-mode for information on the commands to sort the entries and filter them.
			You may want to execute the **pel-compile-info** manually if you want to get information about the current configuration of the PEL compilation control.

Show info about the PEL controlled buffer save management for compile command	**?**	(**pel-compile-info**)	From the specialized ibuffer mode , shown above, open the special help buffer that describes the specialized ibuffer mode shown above. Shows current values and provides buttons to quickly access the customization buffers of the relevant user-options.